Plant Disease Detection System

ITAI 1378 - Computer Vision Midterm Project Proposal

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Tier 1 - Pre-trained Model

The Problem

What: Farmers lose up to 40% of crop yield annually due to plant diseases that go undetected until it's too late.

Who Cares: Farmers, agricultural workers, food supply chain stakeholders

Why Important: Costs billions in lost crops globally. Manual inspection requires expert knowledge and doesn't scale.

Early detection can save crops and reduce pesticide overuse

Our Solution

An image classification system that identifies plant diseases from leaf photos in seconds

Take Photo of Leaf → **Classify Disease** → **Return Diagnosis**

- Enables farmers to diagnose crop issues without expert knowledge
- Uses a pre-trained deep learning model (transfer learning)
- Works with just a smartphone camera

Technical Approach

Component	Choice
Technique	Image Classification
Model	ResNet50 (pre-trained on ImageNet)
Framework	PyTorch + torchvision
Method	Transfer Learning (freeze early layers, fine-tune classifier)

Why ResNet50?

Provides strong classification accuracy out of the box, and transfer learning makes training feasible on free Google Colab GPUs.

Data Plan

Dataset

PlantVillage

(public, open-source)

Size

~54,000

labeled leaf images

Classes

38

across 14 plant species

Labels

Pre-labeled

by disease type

Image Size

224x224

resized + normalized

Split

70/15/15

train / val / test

No manual labeling required - augmentation includes flip, rotate, color jitter

System Architecture

1	2	3	4	5
Camera / Phone Photo	**Preprocessing Resize + Normalize**	**ResNet50 Classification**	**Prediction + Confidence**	**Display to User**
Input	*Preprocessing*	*Model Inference*	*Output*	*Visualization*

End-to-end pipeline: Leaf photo is captured, preprocessed to 224x224, passed through a fine-tuned ResNet50 model, and the predicted disease class with confidence score is displayed to the user.

Success Metrics

Type	Metric	Target
Primary	Classification Accuracy	**>= 90% on test set**
Secondary	Inference Speed	**< 1 second per image**
Additional	Per-class Precision / Recall	**>= 85% for each class**

90%+
Accuracy

<1s
Per Image

85%+
Per Class

Project Timeline

Week	Task	Milestone
10 (Oct 30)	Download PlantVillage dataset, set up Colab environment	Dataset ready
11 (Nov 6)	Load pre-trained ResNet50, train classifier layer	Model working
12 (Nov 13)	Evaluate model, tune hyperparameters, add augmentation	Accuracy >= 90%
13 (Nov 20)	Build Gradio demo app, create demo video	Demo ready
14 (Nov 27)	Final testing, write documentation, clean up repository	Everything done
15 (Dec 4)	**Present project in class**	**Presentation!**

Risks & Mitigation

Risk	Probability	Mitigation (Plan B)
Low accuracy on some disease classes	Medium	Add data augmentation; focus on top 10 most common classes
Colab session timeout during training	Medium	Save model checkpoints after every epoch
Model overfitting on training data	Medium	Use dropout, early stopping, and augmentation
Demo deployment issues	Low	Fall back to Jupyter notebook with sample predictions

Resources

Resource	Details
Compute	Google Colab (free GPU)
Frameworks	PyTorch, torchvision, Gradio
Dataset	PlantVillage (free, open-source)
Estimated Cost	**$0**

Thank You!

Questions?